

S V V Sai Sri Vallabh Pataneni

Internship: Fuel-Cell Stack Industrialisation

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SUMMARY

An Automotive Engineering Master's student at RWTH Aachen specialising in fuel cells and production engineering. At Ecogenium I built and ran a standalone fuel-cell test bench and the surrounding test rigs and fixtures, at DLR I am developing an aircraft fuel-cell system, and at Robert Bosch I worked on fuel-cell operating strategy and analysis. I am seeking the internship in ElringKlinger's fuel-cell industrialisation team to support process optimisation in stack production and the commissioning of new assembly and test equipment near Stuttgart.

KEY SKILLS

Fuel-cell & test: fuel-cell test-bench build & commissioning · polarisation curves · purging · humidity cycling · series testing

Manufacturing: metal 3D printing (DMLS) · DFM/DFAM · production engineering · CATIA V5 · Siemens NX

Simulation: MATLAB/Simulink (MIL) · ANSYS (Structural, ACP)

Programming & data: Python · C/C++ · Power BI

PROFESSIONAL EXPERIENCE

Mechanical / Suspension Team Lead & Pit Crew, Ecogenium e.V (RWTH student team) · 06/2024 – Present

Aachen, Germany

- **Fuel-Cell Test Bench:** Engineered, fabricated and electronically integrated a standalone fuel-cell test bench, running purging, humidity cycling, short-circuiting and polarisation-curve experiments and mapping the stack's optimal thermal operating window.
- **Manufacturing & Test Equipment:** Built and commissioned test rigs and fixtures, designed parts in CATIA V5 / Siemens NX, and prototyped via metal 3D printing (DMLS) with hands-on DFM/DFAM.
- **Energy-Management Strategy:** Architected and deployed energy-management and fuel-cell temperature-control strategies in Python/Simulink (MIL-validated), boosting on-track fuel efficiency by ~80% and securing a runner-up finish at Shell Eco-Marathon 2025.

Werkstudent, Fuel-Cell Systems (Aircraft), DLR Innovation Center for Small Aircraft Technologies (INK) · 01/2026 – Present

Würselen (Aachen), Germany

- **Fuel-Cell System Development:** Developing the ~150 kW fuel-cell system (the engine) for D-LIGHT, DLR's hydrogen-powered 9-seater feeder aircraft.
- **Component Selection:** Selected every component within the thermal-management system against the derived specifications.
- **System Rig Design:** Designed a compact, aircraft-engine-sized fuel-cell system rig in CATIA V5, optimised for easy transferability.

Operating Strategy / System Simulation Intern, Robert Bosch GmbH · 09/2025 – 12/2025

Schwieberdingen, Germany

- **Operating-Strategy Modelling:** Developed and validated an iterative Simulink model of a rule-based power-split operating strategy in the (FC)EVsim PHEV module, correlating outputs against reference Excel and measurement data for high model fidelity.
- **Analysis & Documentation:** Built post-processing and KPI analysis (power distribution, SOC window, efficiency, degradation) and co-authored a peer-reviewed paper on the rule-based FCEV control strategy.

EDUCATION

M.Sc. Automotive Engineering, RWTH Aachen University · 09/2023 – 06/2026

Aachen, Germany · GPA 1.7 (1.2 in core automotive courses) · Fuel-Cell System Technology, Alternative/
Mobile Propulsion, Simulation Sciences, FEM, CFD

B.Tech Mechanical Engineering, JNTU Hyderabad · 06/2019 – 04/2023

Hyderabad, India · GPA 1.3 (German scale)

LANGUAGES

English (Fluent, C1) · German (Intermediate, B1) · Telugu / Hindi (Native)